DECODING MOTOR INTENT FROM SEMG SIGNALS

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Background

Ever since the earliest days of computing, interaction with digital systems around us has remained a challenge. To bridge this gap in interaction, humans have developed and improved technologies that are more expressive and intuitive (Battye et al. 1955). While diverse modalities were developed, which include keyboards, mice and touchscreens, they require constant interaction with a device, which can be limiting (Kaifosh et al. 2025). With the pursuit of developing more natural ways to interact with devices, gesture-based systems that make use of cameras or inertial sensors were developed to avoid an intermediary device. However, these systems faced a major limitation in performance, as they depended only on unobscured movements (Kaifosh et al. 2025). To overcome these limitations, neuromotor interfaces that directly interface with the body's electrical signals were developed to solve the interfacing problem (Hochberg et al. 2012).

These wearable systems could answer the issues of portability and accessibility, while enhancing human computer interaction (HCI). The key motivation to focus on neuromotor signals is the human hand, which is one of the most complex motor systems in the human body, with up to 24 kinematic degrees of freedom (Santello et al. 2013). Enabling HCI with these NMIs, would accomplish a wide range of tasks that are essential for activities of daily living (ADL) (Ting et al. 2021). Consequently, the loss of the hand functions could severely impact a person's ability to live independently (Cappello et al. 2018), making recovery of hand function a top priority for people with tetraplegia after Spinal Cord Injury (SCI) (Ting et al. 2021). Thus, building on these principles, training a decoder pipeline that would effectively enhance the HCI through NMIs, could serve as initial steps to a system that could benefit individuals suffering with motor impairments.

Methods

Participant

1 able-bodied subject participated for 3 sessions in this study and performed 6 different gestures across in each session. The subjects were right-hand dominant and had no history of neuromuscular disorders or injuries. This data was taken from UCI database (Sapsanis et al. 2013).

Surface EMG device and placement

2 Channel EMG device used in this study, which recorded surface EMG signals. It records EMG data at 500 Hz sampling frequency and the channels are placed on flexor carpi ulnaris and extensor carpi radialis. This provides us with enough temporal resolution to perform gesture decoding from the underlying muscle activity in the forearm, thereby establishing our NMI.

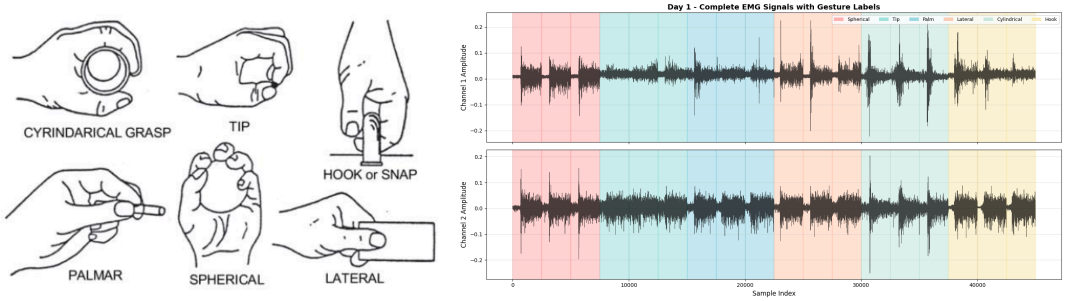


Figure 1: *6 different dextrous gestures being performed and their corresponding EMG activity from a 2 channel device*

Decoder Pipeline

The recorded data was then split into training (80%) and testing (20%) data for each day, to train the models offline either within day or cross day train and testing. Before performing feature extraction and model training, we applied bandpass filter (15Hz to 500Hz) and notch filter (60Hz) to attain the filtered EMG signal. To convert all the negative amplitudes to positive values we perform full-wave rectification of the filtered signal to achieve the rectified signal. A smoothing step

was applied to the rectified signal with the help of a moving root-mean square (RMS) window averaging to attain the smoothed signal.

To perform effective feature extraction, we made use of the filtered, rectified, smoothed signals individually, to assess various features related to intensity and complexity of the EMG activity. We calculated frequency and complexity based features through counting the zero-crossings and slope-sign changes over a 400 milli-seconds window from the filtered and rectified signals respectively. We also calculated other amplitude based time domain features like the root mean square (RMS), mean absolute value (MAV) and wave length (WL) from the smoothed signal over the same window.

A **Features Extracted From Smoothed Signal**

$$\mathrm{RMS}_i = \sqrt{\frac{1}{L}\sum_{n=1}^{L}(x_i[n])^2} \qquad \mathrm{MAV}_i = \frac{1}{L}\sum_{n=1}^{L}|x_i[n]| \qquad \mathrm{WL}_i = \sum_{n=1}^{L-1}|x_i[n+1]-x_i[n]|$$

B **Feature Extracted From Filtered Signal**

$$\mathrm{ZC}_i = \frac{1}{2}\sum_{n=1}^{L-1}|\mathrm{sgn}(x_i[n]) - \mathrm{sgn}(x_i[n+1])|$$

C **Feature Extracted From Rectified Signal**

$$\mathrm{SSC}_i = \sum_{n=2}^{L-1}\mathbb{1}[(x_i[n]-x_i[n-1])(x_i[n]-x_i[n+1])>0 \text{ and } |x_i[n+1]-2x_i[n]+x_i[n-1]|>\theta]$$

Figure 2: *Time-domain EMG features are calculated like (A) RMS, MAV, and WL extracted from smoothed signals. (B) ZC extracted from filtered signals. (C) SSC extracted from rectified signals with threshold θ. Features computed over windows of length L for each channel i.*

We then normalize the features extracted from the EMG recordings in order to prevent any bias to a specific feature or channel and pass them to the PCA module, which performs dimensionality reduction while retaining the principal components that explain 95% variance of the data. This step helps reduce multicollinearity between any highly correlated features and suppresses noise, thereby preserving only the most important principal components. These principal components are subsequently sent to the ML classifiers (LDA / MLP) to decode the intended gestures.

Results

We make use of the decoder pipeline and test the performance of our decoder by training and testing them in 2 situations: 1) within-day training and testing, 2) cross day training and testing, where the data is trained on day 1 and tested on all days. We assessed the offline performance across different types of decoder, starting from a linear model and expanded to non-linear model. We see that the multi-layer perceptron (MLP) performed better in both within (by ~20%) and cross-day cases (~ by 15%). Thereby, showcasing that neural network based models capture more information than linear models.

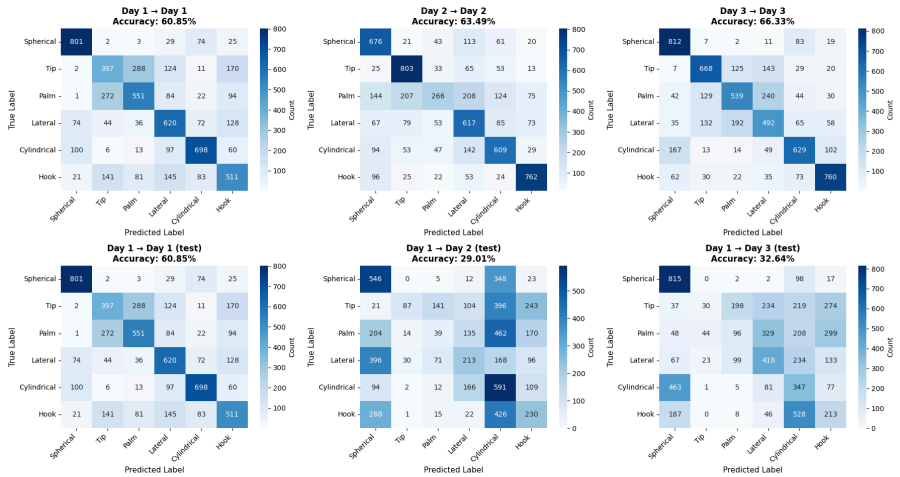


Figure 3: *Confusion matrice for MLP classifier during within and cross day training and testing*

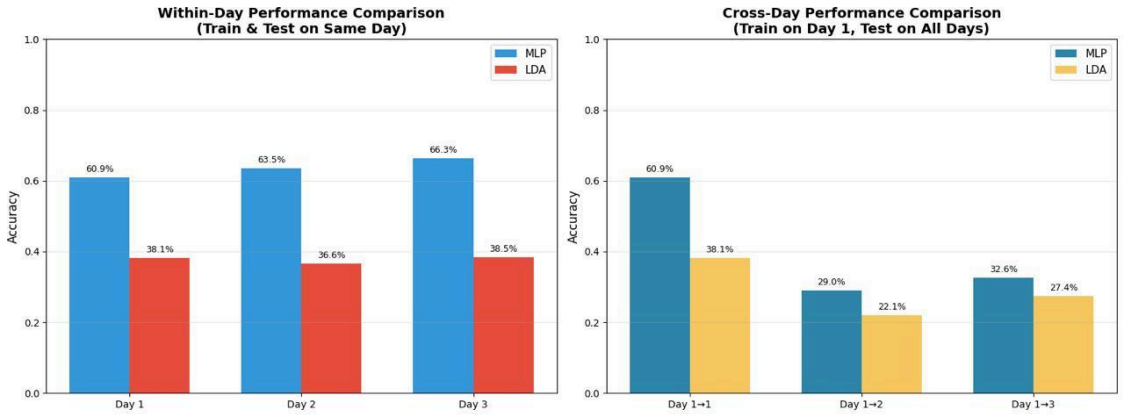


Figure 4: *We compare the performance across LDA and MLP classifier in within and cross day training and testing*

Discussion

In these preliminary results, we found that non-linear classifiers like the MLP outperformed linear classifiers like the LDA. Importantly, we were able to discriminate between each dextrous action performed in the dataset using the MLP classifier with approximately 60% accuracy when calibrated everyday, but see a decrease in performance similar to the LDA model in cross day training and testing. Even though we see a decrease, the MLP classifier still performed better than the LDA model, showing better pattern recognition ability. The reason why we see decrease in cross day performance could be due to electrode shifts, only 2 channels capturing dextrous gestures, inter-day variability due to skin conductances or temporal data drift.

Although preliminary, these results provide further insights into improving HCI with the help of EMG signals. In future work, we hope to make use of better classifiers like the CNN or a transfer learning based approach to improve performance of decoding between multiple gestures, to allow multiple degrees of freedom of control. We could also include multiple channels to capture more data from multiple muscles to understand the complex data produced by dextrous movements

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